

ROS 2 Engineering Handbook

Beginner to Advanced

Study Edition

May 29, 2026

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Preface

This book provides a structured path from ROS 2 fundamentals to advanced middleware, DDS, executors, lifecycle nodes, and production robotics systems.

Part I

Foundations

Chapter 1

What is ROS 2?

ROS 2 is a distributed robotics middleware framework built on DDS.

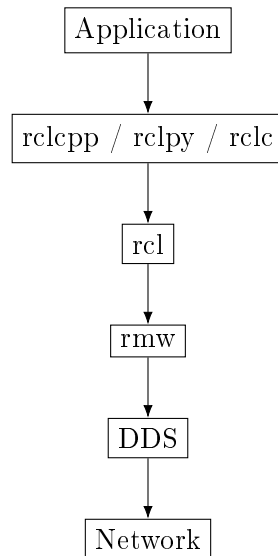
1.1 Major Concepts

- Nodes
- Topics
- Services
- Actions
- Parameters
- DDS Middleware

Chapter 2

ROS 2 Architecture

2.1 Layered Architecture



Chapter 3

Workspace and Build System

3.1 Workspace Structure

```
workspace/  
|-- src/  
|-- build/  
|-- install/  
'-- log/
```

3.2 Build

```
colcon build  
source install/setup.bash
```


Part II

Programming with ROS 2

Chapter 4

C Development (rclc)

4.1 Embedded Architecture

```
Application
|
rclc
|
rcl
|
rmw
|
DDS
```

Chapter 5

C++ Development (rclcpp)

5.1 Publisher Example

```
class Talker : public rclcpp::Node
{
public:
    Talker() : Node("talker")
    {
        publisher_ =
            create_publisher<std_msgs::msg::String>(
                "chatter", 10);
    }
};
```

Chapter 6

Python Development (rclpy)

6.1 Publisher Example

```
class Talker(Node):  
  
    def __init__(self):  
        super().__init__("talker")  
  
        self.pub = self.create_publisher(  
            String,  
            "chatter",  
            10  
        )
```

Part III

Communication

Chapter 7

Topics



Chapter 8

Services

Client --> Request
Server --> Response

Chapter 9

Actions

Goal

|

Feedback

|

Result

Chapter 10

QoS

Policy	Description
Reliability	Reliable / Best Effort
Durability	Message persistence
History	Queue depth
Deadline	Timing guarantees

Part IV

DDS Deep Dive

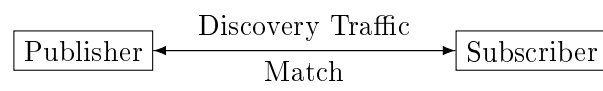
Chapter 11

DDS Entities

- Domain Participant
- Publisher
- Subscriber
- DataWriter
- DataReader

Chapter 12

Discovery



Part V

Talker and Listener Internals

Chapter 13

Process Creation

Command:

```
ros2 run demo_nodes_cpp talker
```

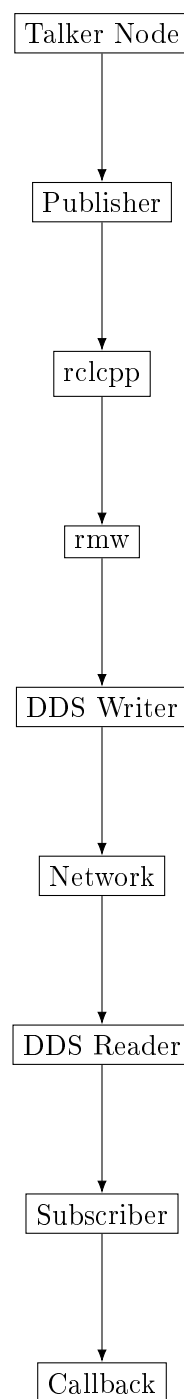
Operating system creates:

PID 1001

Talker Process

Chapter 14

Message Journey



14.1 Detailed Flow

1. OS creates process.
2. ROS initializes.
3. Node created.
4. Publisher created.
5. DDS participant created.
6. Discovery begins.
7. Subscriber matches topic.
8. Message serialized.
9. DDS transmits bytes.
10. Subscriber receives bytes.
11. Message deserialized.
12. Executor schedules callback.
13. User callback executes.

Chapter 15

Executors

15.1 Single Threaded

```
rclcpp::spin(node);
```

Responsibilities:

- Wait for DDS events
- Execute callbacks
- Process timers
- Handle services and actions

Part VI

Advanced ROS 2

Chapter 16

Lifecycle Nodes

```
Unconfigured
|
Configured
|
Active
|
Inactive
|
Finalized
```

Chapter 17

Composition

Multiple nodes can run inside a single process.

Benefits:

- Less memory
- Faster communication
- Reduced context switching

Chapter 18

Real-Time ROS 2

Topics:

- Deterministic scheduling
- Memory pre-allocation
- Lock-free structures
- Real-time executors

Chapter 19

Security

SROS2 provides:

- Authentication
- Encryption
- Access control

Part VII

Robot Systems

Chapter 20

Navigation Stack

Map
|
Localization
|
Planning
|
Controller
|
Robot

Chapter 21

Perception

Camera, LiDAR, IMU and sensor fusion systems.

Chapter 22

Manipulation

MoveIt 2 architecture and robot arm planning.

Part VIII

Projects

Chapter 23

Suggested Projects

1. Differential drive robot
2. SLAM robot
3. Warehouse robot
4. Multi-robot fleet
5. Autonomous rover

Further Reading

Official ROS 2 documentation, Navigation2, MoveIt 2, Gazebo and Micro-ROS.